



ECS 2026

Maximizing Efficiency

A Deep Dive into Azure VM SKU Optimizations

Gregor Reimling

ALL OF YOUR AWESOME TITLES AND
AWARDS THAT YOU NEED TO SHOW
TO THE WORLD





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Azure & Security





PREMIUM



PREMIUM PARTNER



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DIAMOND



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PLATINUM



GOLD



SILVER



BRONZE



FUTURE MAKER SILVER



FUTURE MAKER BRONZE



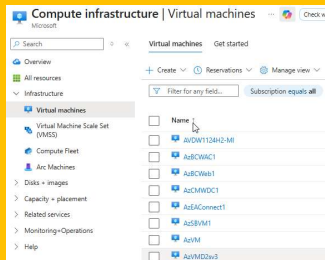
MEDIA PARTNER



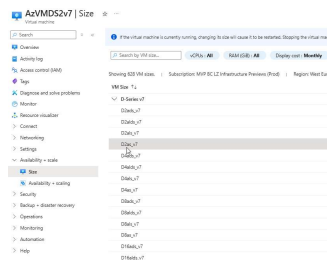


Agenda

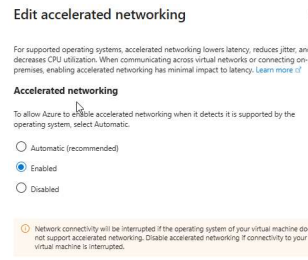
VM Optimization with Azure Boost



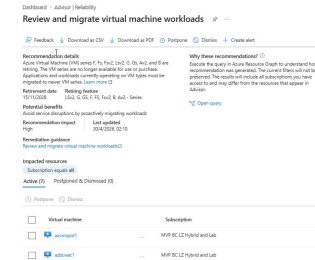
Azure VM
Overview



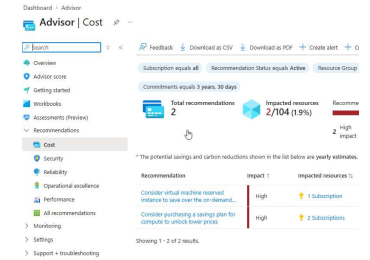
Azure Boost
Architecture



Introduction v6
and v7 SKUs



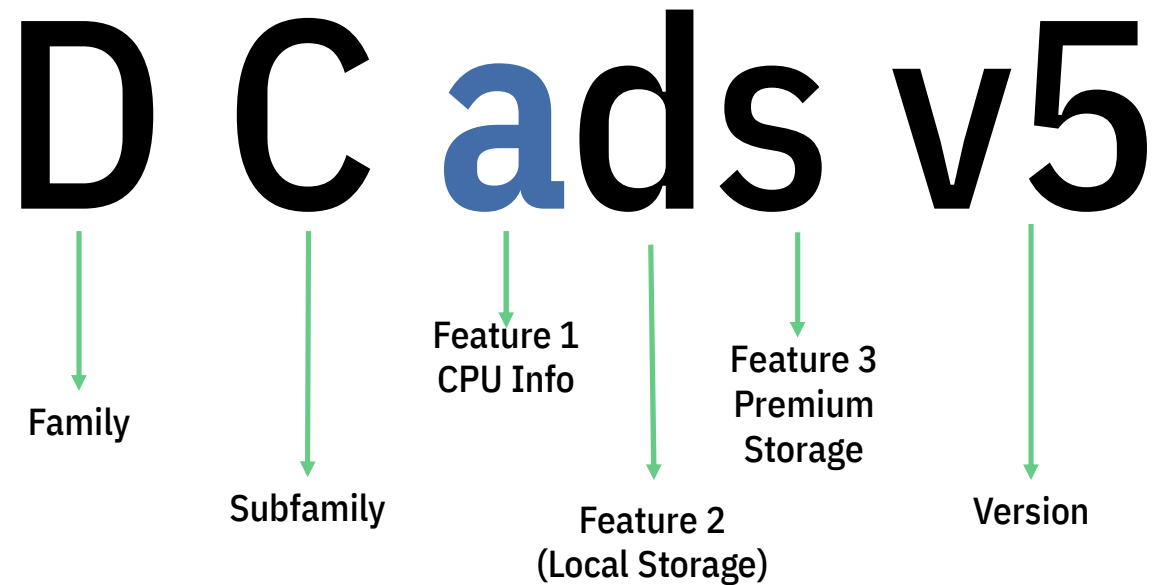
Migration paths
& Planning



Pricing and
Takeaways



Azure VM SKU Naming



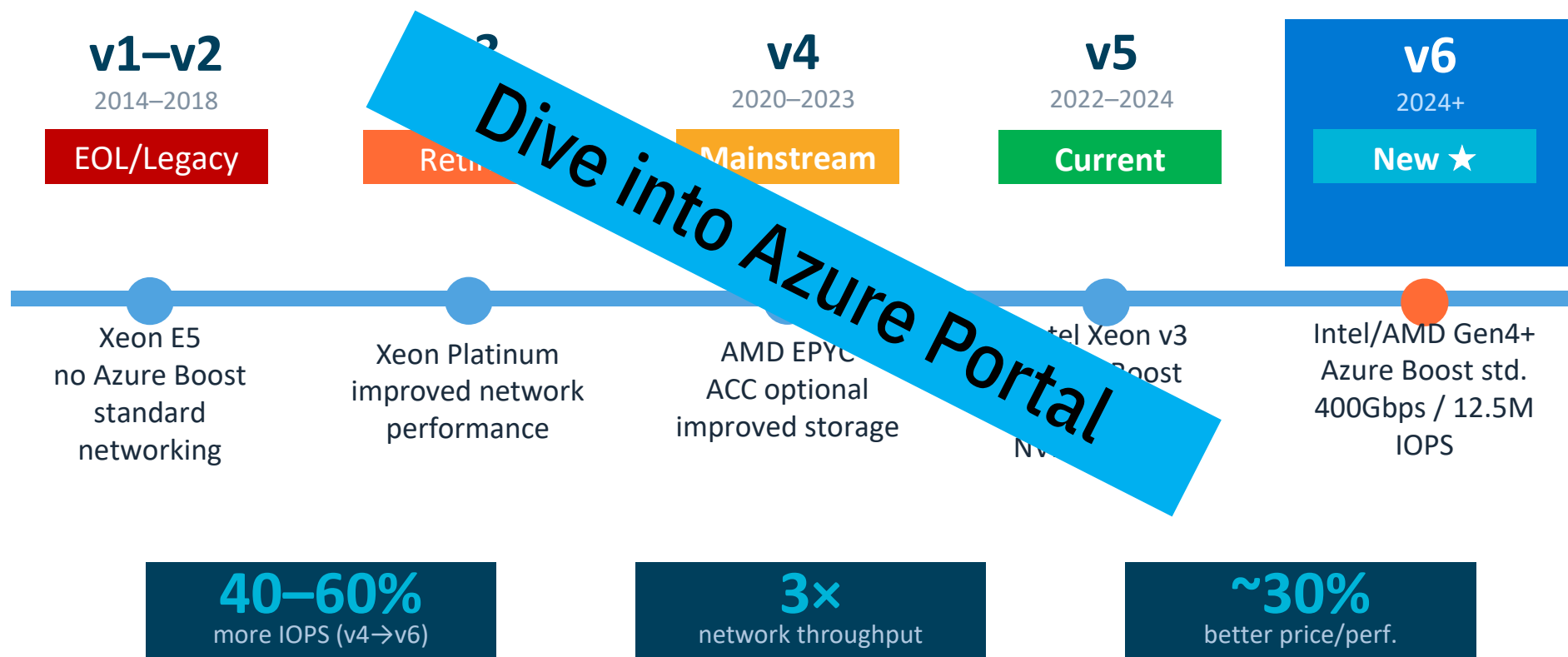
No letter = Intel

a = AMD

p = ARM



SKU Generations at a Glance





Retired VMs

Series name	Retirement Status	Retirement Announcement	Planned Retirement Date	Migration Guide
D-series	Announced	<u>03/31/25</u>	05/01/28	-
Ds-series	Announced	<u>03/31/25</u>	05/01/28	-
Dv2-series	Announced	<u>03/31/25</u>	05/01/28	-
Dsv2-series	Announced	<u>03/31/25</u>	05/01/28	-

Retired Azure VM size series - Azure Virtual Machines

Woodler



Why VM Optimization?



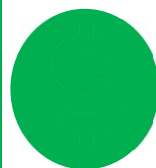
SKU Lifecycle

Old SKUs (v1–v4) are reaching end-of-life. Microsoft recommends migrating to newer generations.



Performance Gap

Legacy VMs cannot meet new workload demands – especially for I/O-intensive applications.



Price-to-Performance

New SKUs offer up to 40–50% more performance at the same or lower cost thanks to modern hardware and lower CO2 emissions.



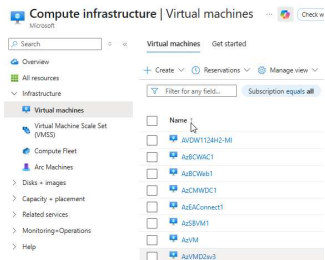
Security & Compliance

Newer SKUs offer enhanced security features: vTPM, Secure Boot, and Confidential Computing.

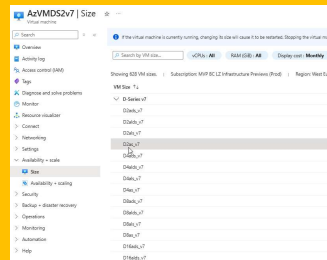


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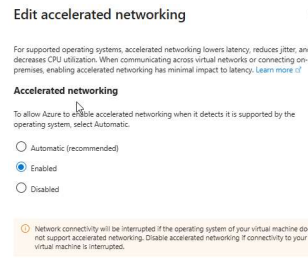
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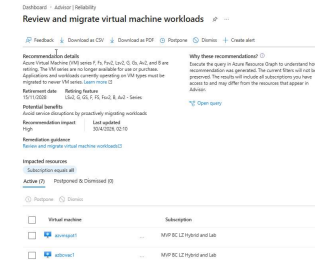
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Overview



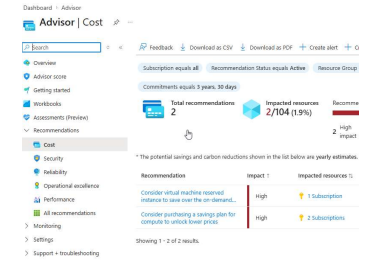
Azure Boost
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Introduction v6
and v7 SKUs



Migration paths
& Planning



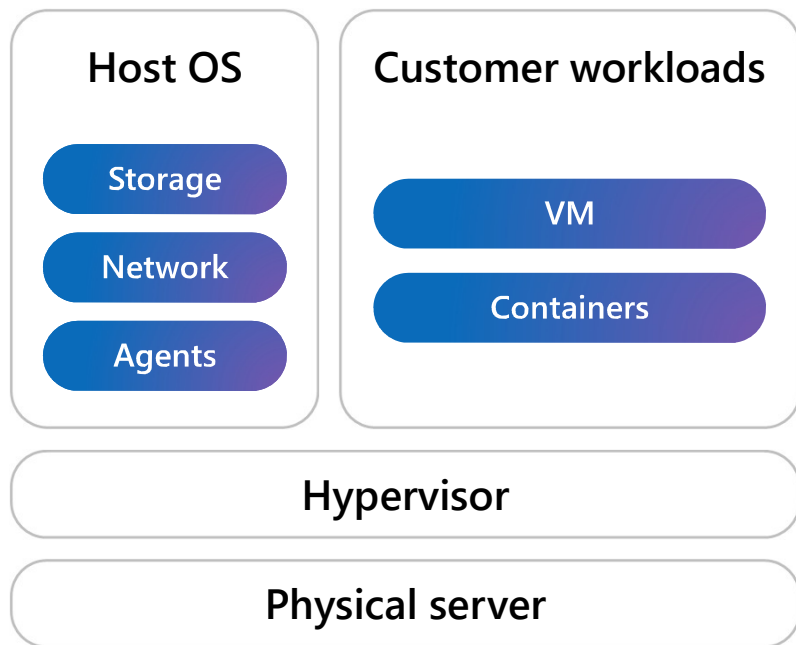
Pricing and
Takeaways



Infrastructure acceleration through offload

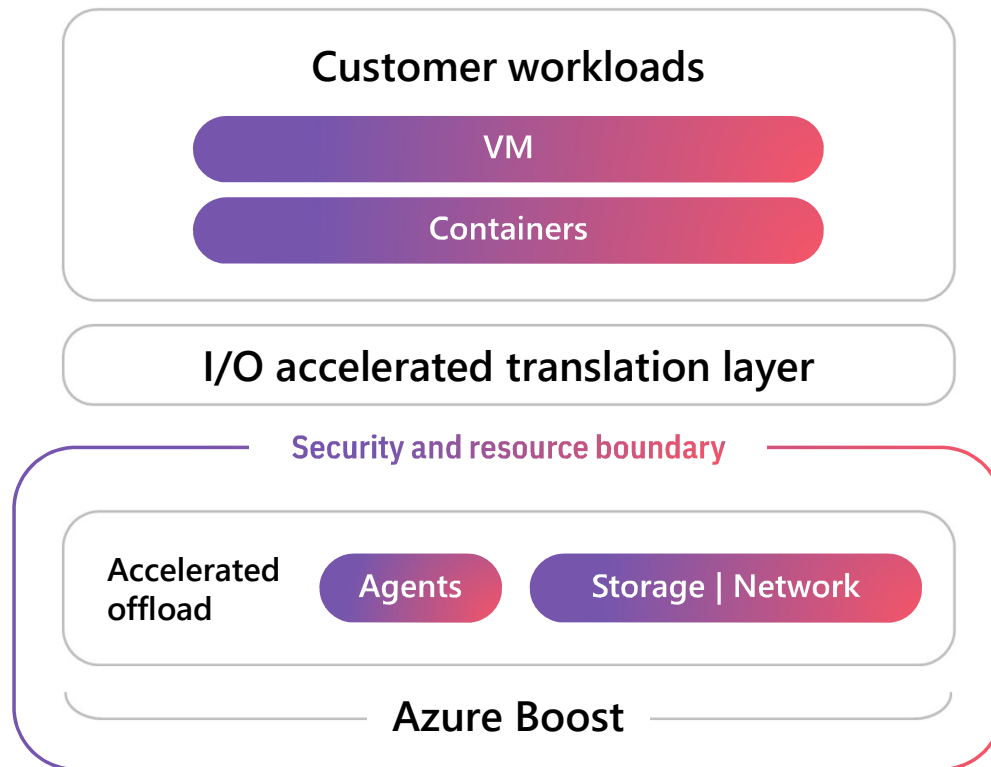
Traditional infrastructure

Network and storage I/O processed in Host OS



Offloaded infrastructure

Server is managed through dedicated offload card





Azure Boost benefits

Storage

Remote

20 GB/s

1.4x increase from Build '25
Azure Managed Disks throughput

1M

1.6x increase from Build '25
Azure Managed Disks IOPS

Local

36 GB/s

Azure Boost SSD
throughput

6.6M

Azure Boost SSD
IOPS

Networking

400* Gb/s

2x increase from Build 25'
network throughput for GP and AI SKU

400K

Connections per
second

Security

**Azure Boost Confidential
Device**

TDISP Support

Maintenance

Dual TOR

Active Active networking
support

Sub-second

networking maintenance

Deployment

>25%

1.5x increase from Build '25
Azure fleet is Boost enabled

100%

New SKUs ship with
Azure Boost

GPU and HPC

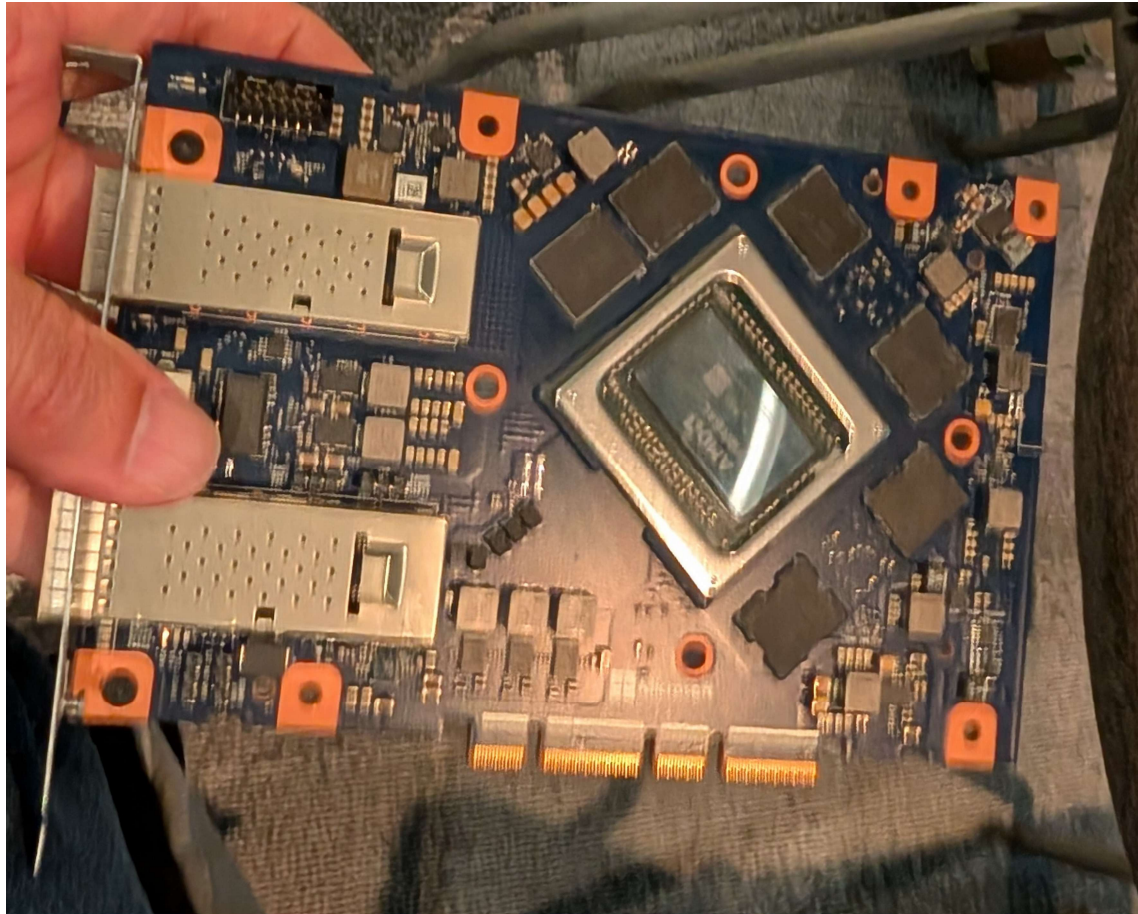
Support

Physical and memory isolation between Azure Boost and customer workloads

* With speeds up to this number.



Picture of Azure Boost Add-In Card

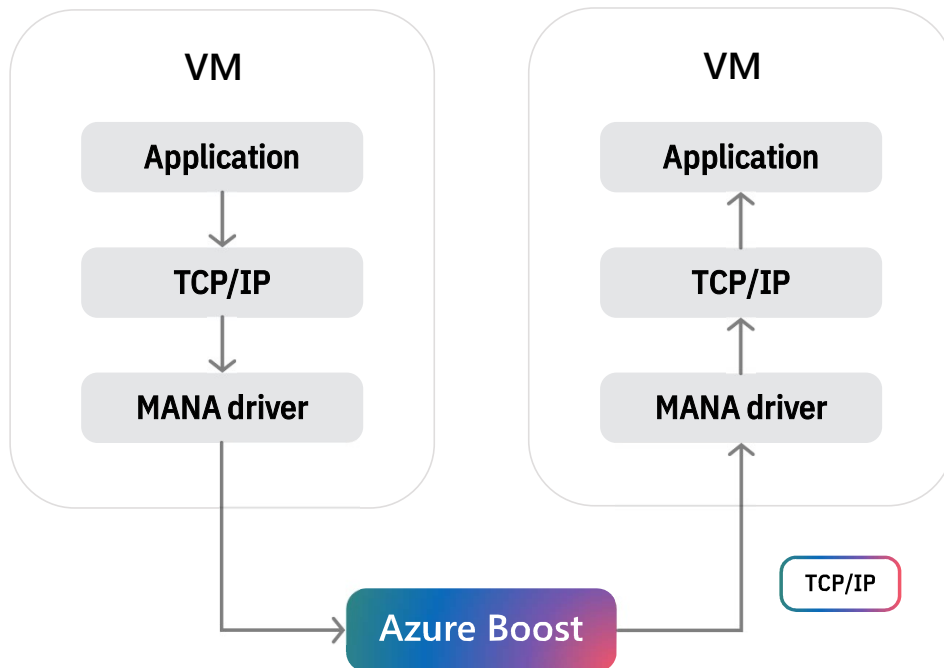




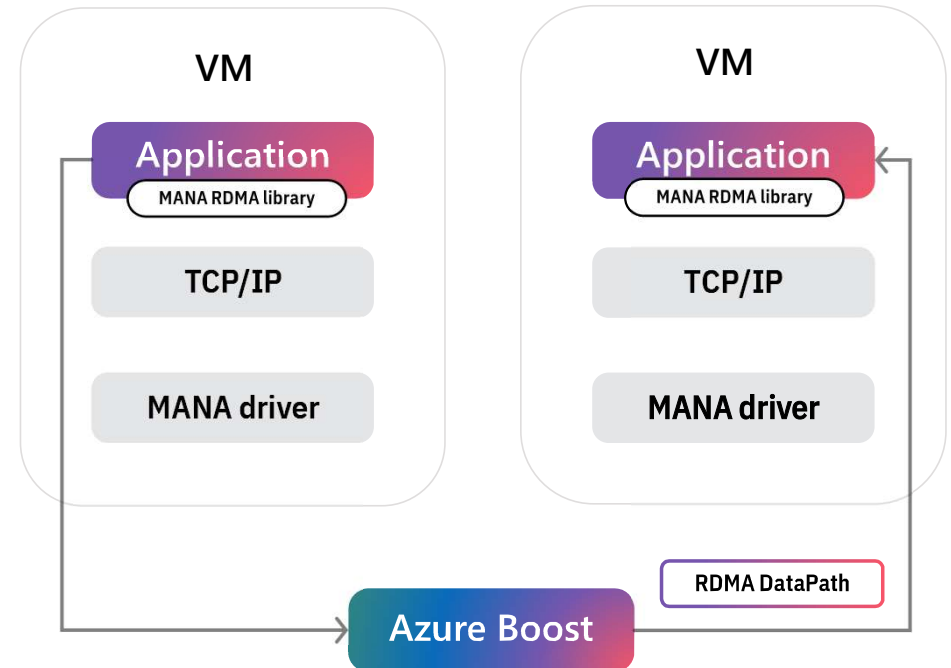
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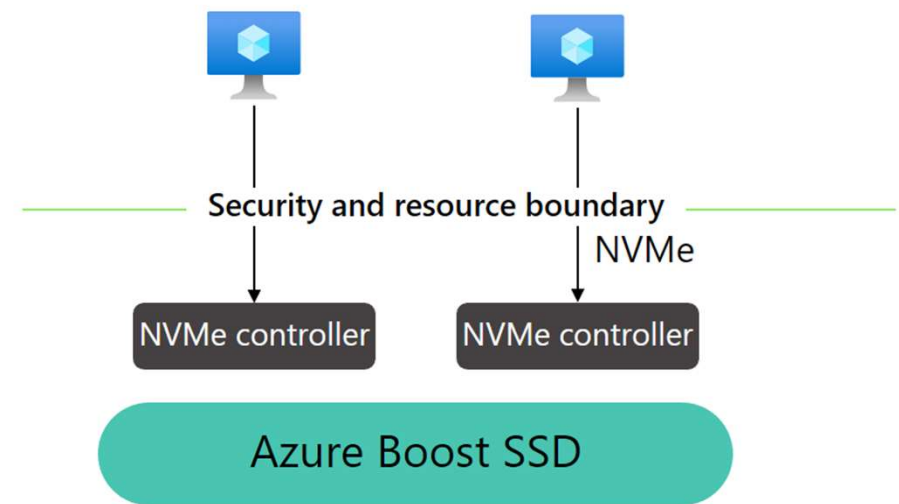
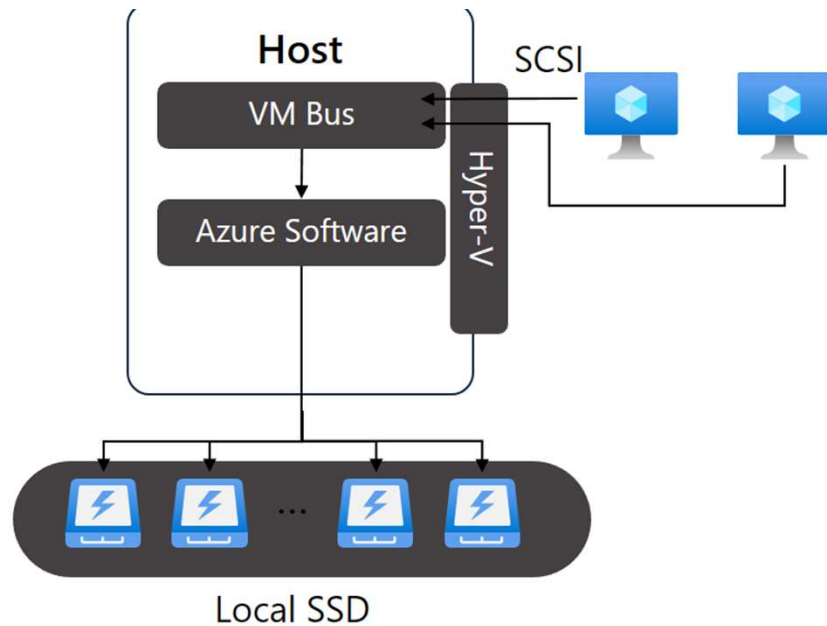
Remote Direct Memory Access (RDMA)

TCP/IP



RDMA





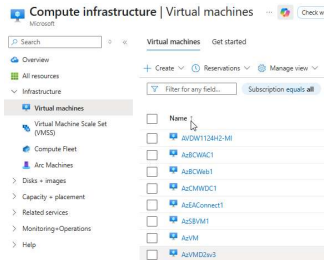
Disk Performanceboost SCSI 2 NVMe

- NVMe provides significantly higher performance compared to SCSI
- Enabled through a new disk driver and PGA
- NVMe support is only available starting with v6 and in selected v5 SKUs
- NVMe is supported **only on Gen2** virtual machines

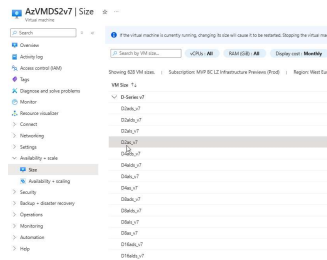


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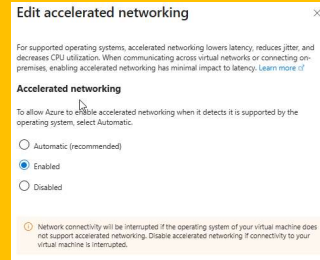
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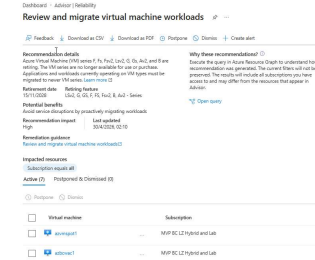
Azure VM Overview



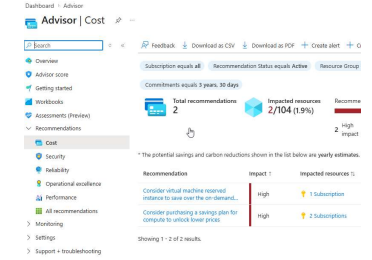
Azure Boost Architecture



Introduction v6 and v7 SKUs



Migration paths & Planning

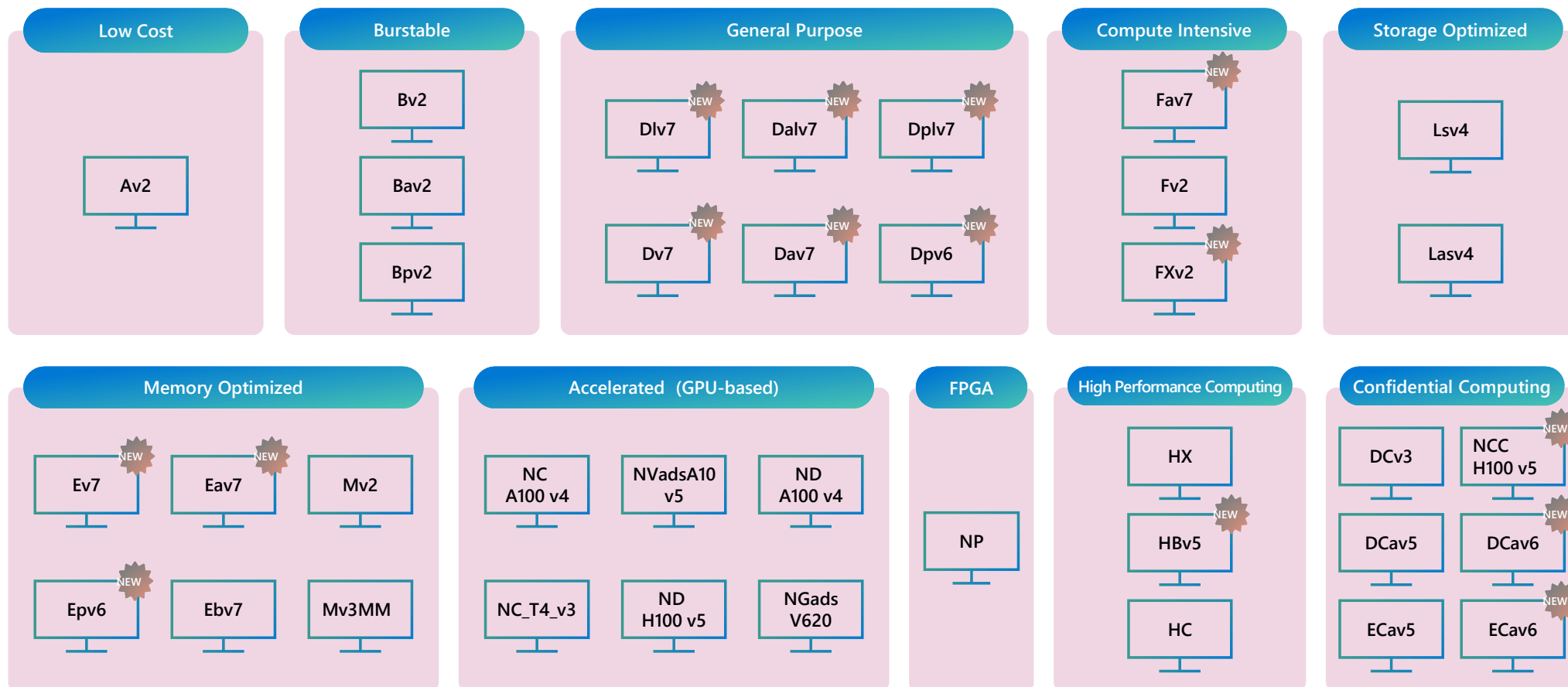


Pricing and Takeaways



as of 02/2026

Azure Compute (VM) portfolio

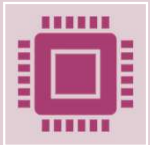


More details on these and prior versions of VMs can be found at <https://azure.microsoft.com/en-us/pricing/details/virtual-machines/series/>

Azure Region Lookup: Instance Types, VM Series/Sizes



Azure v6 SKUs and above



Increased CPU Cache: Up to 5× more cache (from 60 MB to 300 MB) dramatically improves data access speeds.



NVMe for Storage: Enhanced local and remote storage performance, with up to 3× more IOPS locally and the capability to reach 400k IOPS remotely via Azure Boost.



Azure Boost: Delivers higher throughput (up to 12 GB/s remote disk throughput) and improved network bandwidth (up to 200 Gbps for larger sizes).



Microsoft Azure Network Adaptor (MANA): Provides improved network stability and performance for both Windows and Linux environments.



Azure Cobalt 100 Arm-based VMs

General Purpose (lower memory to vCPU ratio)	General Purpose	Memory Optimized
		
Dpls and Dplds v6	Dps and Dpds v6	Eps and Epds v6
Up to 192 GiBs of memory (2 GiBs per vCPU)	Up to 384 GiBs of memory (4 GiBs per vCPU)	Up to 672 GiBs of memory (8 GiBs per vCPU)
Up to: 96 vCPUs 5280 GiBs of local storage 60 Gbps of network bandwidth 40 Gbps of remote storage throughput		
Workloads		
Media encoding, gaming servers, small databases, microservices	Application servers, web servers, small to medium databases	Large databases, in-memory caches, and analytics workloads

Ability to attach Standard, Premium and Premium SSD v2, and Ultra Disks (based on regional availability)



Local Temp Disk vs Remote Temp Disk

v6 VMs include raw local temporary disks backed by NVMe

Remote temporary disks are provisioned and unformatted after VM deployment

Data stored on NVMe-backed remote temporary disks persists across reboots

Persistence lasts until VM redeployment or host maintenance

The page file is located on the OS disk by default

Dive into Azure Portal



Microsoft Azure Network Adapter (MANA)



is a core component of Azure Boost



Provides increased network
performance and improved availability

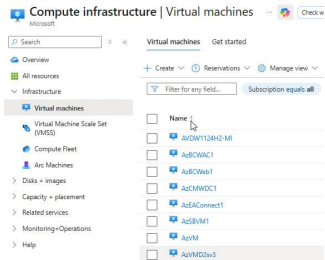


Supported starting with Windows
Server 2019

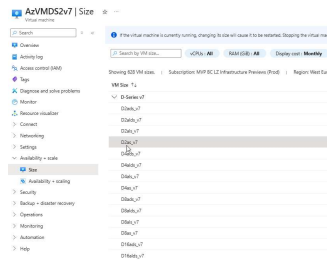


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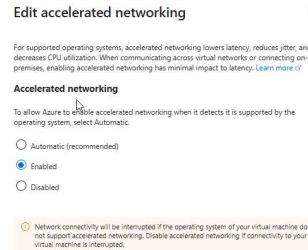
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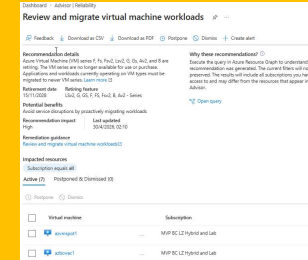
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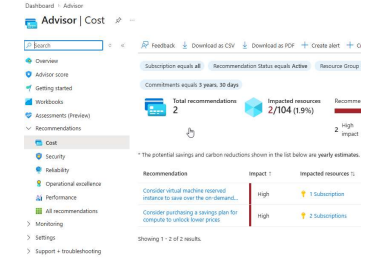
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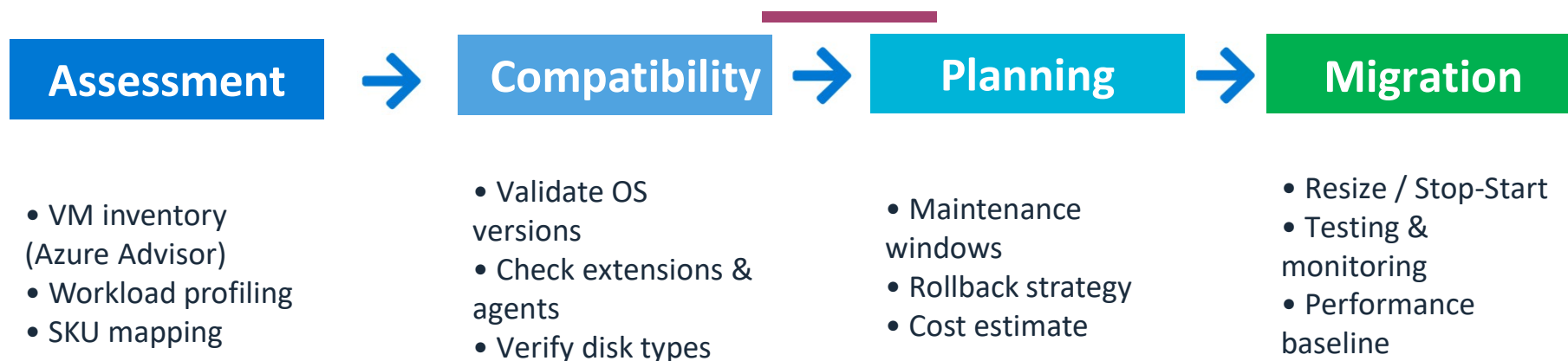
Migration paths & Planning



Pricing and Takeaways



Migration Paths



Recommended Migration Path

v3 → v5: Direct resize usually possible. Check storage type (Standard → Premium SSD v2).

v4 → v6: Recommended: Stop-Deallocate + Resize. Watch the disk controller type (SCSI → NVMe).

All: Optimize OS Disk caching settings for NVMe. Gen2 VM required for all v6 SKUs.



Challenges & Backward Compatibility

NVMe instead of SCSI

v6 SKUs use an NVMe disk controller. Older Windows images may not boot from NVMe.

Gen2 VM Required

Azure Boost requires Generation 2 VMs. Migration from Gen1 requires an image conversion.

SCSI Drivers

Some older Linux distributions need manual NVMe driver updates before migration.

Ultra Disk Zones

Ultra Disk is only available in specific regions and availability zones.



Use new OS images with NVMe drivers. Windows Server 2019+ and modern Linux



Use Azure Migrate Assessment to check Gen1/Gen2 status. Convert or redeploy with



Use recent kernel versions (Ubuntu 20.04+, RHEL 8.4+). Test drivers in Dev/Test



Check availability zone and region support before planning: `az disk list-skus`.



Migration Paths

Azure Advisor

Assessment

Automated SKU recommendations based on utilization metrics. Detects under/over-provisioned VMs.

Azure Migrate

Assessment

Discovery, performance pre-migration, planning for Azure-native infrastructure.

Azure Cost Mgmt.

Planning

What-if analysis, budgets and Reserved Instance recommendations after migration.

VM Resize (Portal/CLI)

Migration

Direct resize while running (same cluster family). CLI: `az vm resize` – fastest path.

Azure Monitor

Monitoring

Post-migration baseline: compare CPU, IOPS, network throughput and latency metrics.

Azure Policy

Enforcement

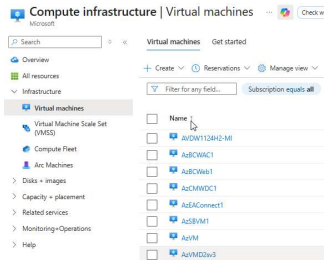
Allowed SKU lists and block deprecated SKU families – ensures compliance at scale.

Dive into Azure Portal

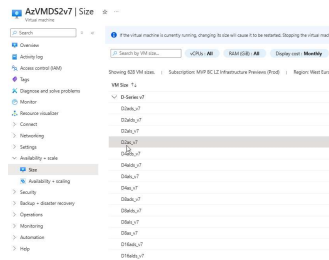


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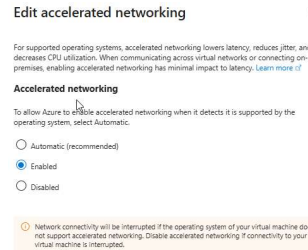
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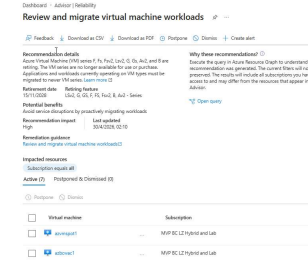
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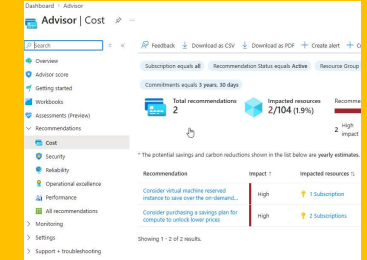
Azure Boost Architecture



Introduction v6 and v7 SKUs



Migration paths & Planning



Pricing and Takeaways



Price/Performance Overview

~30%

Cost savings
v4→v6 (Pay-as-you-go)

2×

More vCPUs at
the same price

40%

Fewer VMs needed
through higher density

Example Comparison: 50 VMs D4s_v3 → D4ds_v6

Metric	D4s_v3 (current)	D4ds_v6 (new)	Delta
vCPUs	4	4	same
RAM	16 GB	16 GB	same
Max IOPS	6,400	150,000	+2,244%
Max Network	3.2 Gbps	12.5 Gbps	+290%
Price/hr (West Europe)	~\$0.192	~\$0.155	-19%



Key Takeaways

01

Azure Boost is not just a feature – it's a platform shift. I/O offloading at the hardware level fundamentally redefines VM performance.

02

v6 SKUs offer the best price-to-performance ratio in Azure history. Migrations often pay back within 6 months.

03

Backward compatibility is not a showstopper – but it requires planning: NVMe drivers, Gen2 VMs and disk types must be validated upfront.

04

Regular SKU reviews every 12–18 months are a strategic necessity for infrastructure-intensive workloads, not optional.

05

Start with Azure Advisor + Azure Migrate – both tools provide actionable SKU recommendations and estimated cost savings directly in the portal.



SQL Server on Azure VM v6



Einsatz von V6 SKUs erhöht Performance deutlich

NVMe Protokoll bringt deutlich bessere IOPS Werte

Migration durch Änderung der SKU nicht möglich – nur durch Lift and Shift

Local Temp Disk Einsatz nicht supported

Extra Disk für Temp Disk anlegen und VMs mit Temp Disk vermeiden



Links

- [Virtual machine sizes overview - Azure Virtual Machines](#)
- [Retired Azure VM size series - Azure Virtual Machines](#)
- [Virtual machine sizes overview - Azure Virtual Machines](#)
- [Azure Region Lookup: Instance Types, VM Series/Sizes](#)
- [Microsoft Azure Network Adapter \(MANA\) overview](#)
- [NVMe overview - Azure Virtual Machines](#)

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